

At the end of this worksheet you should be able to

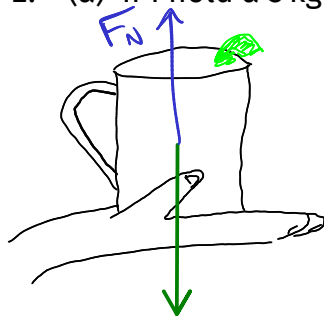
- discuss Newton's laws and provide examples of the application of each.
- apply Newton's first law to solve interesting physical problems.
- apply Newton's second law to solve interesting physical problems for objects that accelerate.
- apply Newton's third law to situations involving the motion of multiple objects.

1. What base units are the composite force units of Newtons equal to?

$$\sum \vec{F} = m \cdot \vec{a}$$

$$[\text{Newtons}] = [\text{kg}] \cdot \left[\frac{\text{m}}{\text{s}^2} \right] = \left[\frac{\text{kgm}}{\text{s}^2} \right]$$

2. (a) If I hold a 5 kg cup motionless in my hand, what force do I provide to the cup?



$$5 \text{ kg} \cdot 9.8 \frac{\text{N}}{\text{kg}} = 49 \text{ N}$$

	x	y
F_g	0	-49N
F_N	0	F_N
ΣF	0	0

$$\underline{F_N = 49 \text{ N}}$$

(b) If I raise a 5 kg cup with my hand, at *constant speed*, then what force do I need to provide from my hand to the cup?



	x	y
F_g	0	-49N
F_N	0	F_N
ΣF	0	0

$$\underline{F_N = 49 \text{ N}}$$

(c) If I accelerate the cup upwards with an acceleration of $+1 \text{ m/s}^2$, then what force does my hand need to provide to the cup?

$$\Sigma F_y = ma_y$$

$$\Sigma F_y = 5 \text{ kg} \cdot 1 \frac{\text{m}}{\text{s}^2}$$

$$\Sigma F_y = 5 \text{ N}$$

	x	y
F_g	0	-49N
F_N	0	F_N
ΣF	0	+5N

$$-49 \text{ N} + F_N = 5 \text{ N}$$

$$\underline{\underline{F_N = 54 \text{ N}}}$$

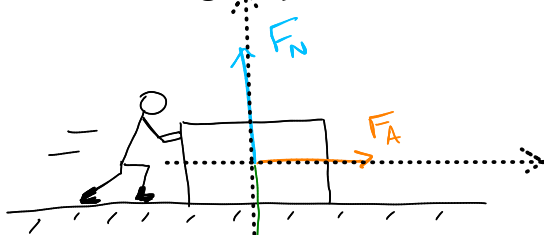
- (d) If I accelerate the cup downwards with an acceleration of -1 m/s^2 , then what force does my hand need to provide to the cup?

$$\begin{aligned}\Sigma F_y &= ma_y \\ \Sigma F_y &= 5 \text{ kg} \cdot (-1 \text{ m/s}^2) \\ \Sigma F_y &= -5 \text{ N}\end{aligned}$$

	x	y
F_g	0	-49 N
F_N	0	F_N
ΣF	0	-5 N

$$\begin{aligned}-49 \text{ N} + F_N &= -5 \\ F_N &= +44 \text{ N}\end{aligned}$$

3. If I push a 100 kg box with an applied force of 700 N along a friction-less surface. Find the force of gravity on the box, the normal force, and the net force.



$$\begin{aligned}F_g &= 100 \text{ kg} \cdot 9.8 \frac{\text{N}}{\text{kg}} \\ F_g &= 980 \text{ N}\end{aligned}$$

Bonus: What is the acceleration?

$$\begin{aligned}\Sigma F_x &= ma_x \\ +700 \text{ N} &= 100 \text{ kg} \cdot a_x \Rightarrow a_x = +7 \text{ m/s}^2\end{aligned}$$

	x	y
F_g	0	-980
F_N	0	F_N
F_A	+700 N	0
ΣF	+700 N	0 N

vector addition

$a_y = 0 \therefore \Sigma F_y = 0$
"observation"

$$\begin{aligned}-980 + F_N + 0 &= 0 \\ F_N &= 980 \text{ N}\end{aligned}$$

4. Take the same problem from above and now add friction. The coefficient of friction is $\mu = 0.5$ then what is the net force?

$$\mu = \frac{|F_f|}{|F_N|}$$



$$0.5 = \frac{F_f}{980 \text{ N}}$$

$$F_f = 490 \text{ N}$$

$$\begin{aligned}F_g &= 100 \text{ kg} \cdot 9.8 \frac{\text{N}}{\text{kg}} \\ F_g &= 980 \text{ N}\end{aligned}$$

	x	y
F_g	0	-980
F_N	0	F_N
F_A	+700 N	0
F_f	-490 N	0
ΣF	+210 N	0

$$F_N = 980 \text{ N}$$

What is the acceleration? $210 \text{ N} = 100 \text{ kg} \cdot a_x$

$$a_x = 2.1 \text{ m/s}^2 \checkmark$$

5. For both of the previous problem, what happens when I stop pushing?

#3: frictionless

$$a_x \rightarrow 0 \text{ m/s}^2$$

#4: friction full

slow down

X	Y
F_g	0
F_N	0
F_A	0
F_f	-490 N
ΣF	-490

$$a_x = -4.9 \text{ m/s}^2$$

6. What force do I need to exert to push the box at constant speed?

#3: frictionless

$$F_A = 0 \text{ N}$$

#4: friction full

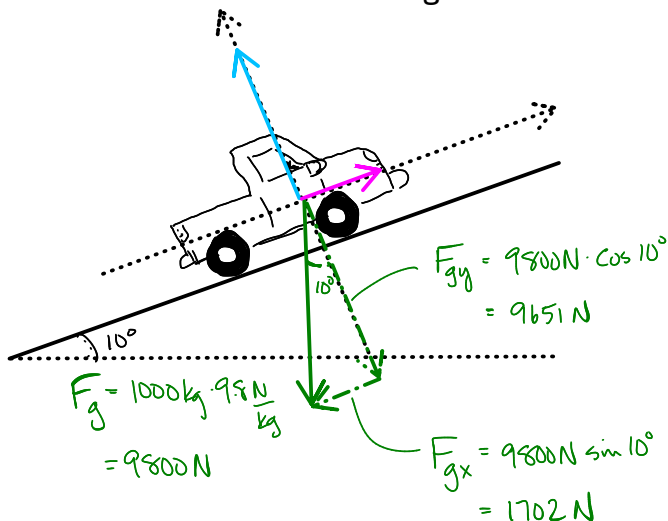
$$F_A = +490 \text{ N}$$

X	Y
F_g	0
F_N	0
F_A	+490
F_f	-490 N
ΣF	0

$$a_x = 0$$

7. If I need to provide a 1000 N force to keep a 100 kg box moving at constant speed along a level floor, then what is the coefficient of friction between the floor and the box?

8. A 1000 kg car is parked on a hill that has an angle of 10° with respect to the horizontal. What is the weight of the car? What is the normal force on the car? What force is keeping the car from sliding down the hill? How large is that force?



X	Y
F_g	-1702
F_N	0
F_f	F_f
ΣF	0

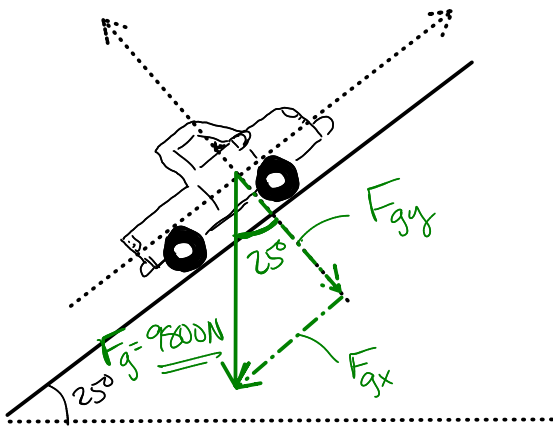
$$-1702 + F_f = 0$$

$$F_f = 1702 \text{ N}$$

$$-9651 + F_N = 0$$

$$F_N = 9651 \text{ N}$$

9. If the maximum incline that the car can be parked on without sliding is 25° , then what is the coefficient of friction between the tires and the road?



	x	y
F_g	-4142	-8882
F_N	0	F_N
F_f	F_f	0
ΣF	0	0

$$F_N = 8882$$

$$F_f = 4142$$

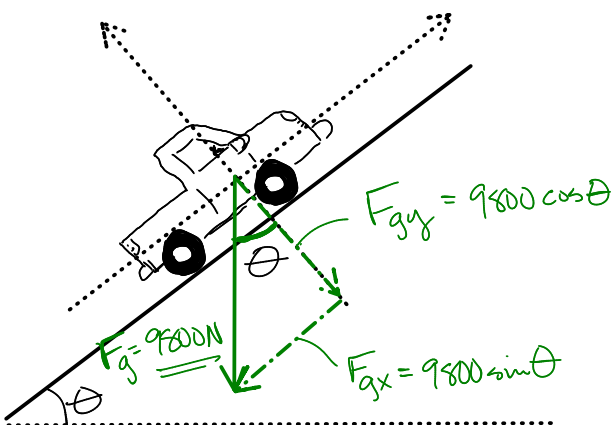
$$\mu_s = \frac{|F_f|}{|F_N|} = \frac{4142}{8882} = 0.47$$

10. Work the previous problem inside-out.

$$\mu = 0.47$$

$$\theta = ?$$

if a ~~1000 kg~~ car is parked on an incline with a $\mu = 0.47$, what is the steepest incline it can be parked on and not slide?



	x	y
F_g	$-9800 \sin \theta$	$-9800 \cos \theta$
F_N	0	F_N
F_f	F_f	0
	0	0

$$-9800 \cos \theta + F_N = 0$$

$$F_N = 9800 \cos \theta$$

$$-9800 \sin \theta + F_f = 0$$

$$F_f = 9800 \sin \theta$$

$$\mu = \frac{F_f}{F_N}$$

$$0.47 = \frac{9800 \sin \theta}{9800 \cos \theta}$$

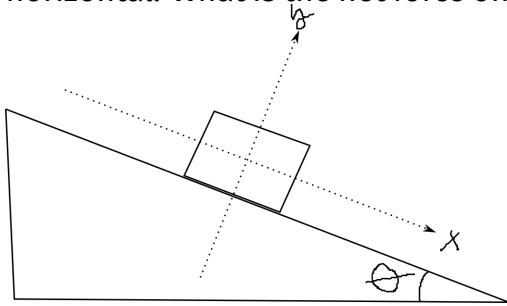
$$0.47 = \frac{\sin \theta}{\cos \theta}$$

$$0.47 = \tan \theta$$

$$\theta = \tan^{-1}(0.47)$$

$$\theta = 25^\circ$$

11. A 100 kg box slides down a friction-less inclined plane that has an angle of 30° to the horizontal. What is the net force on the box and then what is the acceleration of the box?

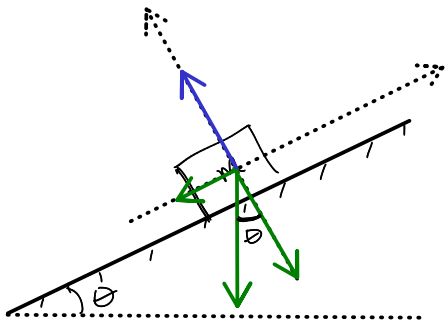


$$a = 4.90 \text{ m/s}^2$$

12. A 100 kg box slides down a friction-full inclined plane that has an angle of 30° to the horizontal and a coefficient of friction of $\mu = 0.1$. What is the net force on the box and then what is the acceleration of the box?

$$a_x = 4.05 \text{ m/s}^2$$

13. Let's do the friction-less inclined plane problem *in general* for any mass and incline. Follow the same procedure as before but with the variable m for mass and θ for incline angle. Find an expression for the net force on the mass as a function of θ and for the acceleration as a function of θ .



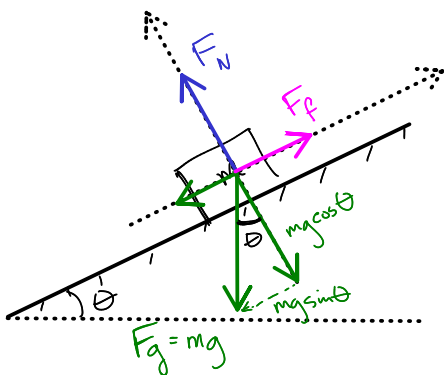
	x	y
F_g	$-mg \sin \theta$	$-mg \cos \theta$
F_N	0	F_N
ΣF	$-mg \sin \theta$	0

$$ma_x = \Sigma F_x = -mg \sin \theta \quad F_N = mg \cos \theta$$

$$\frac{ma_x}{m} = \frac{-mg \sin \theta}{m}$$

$$a_x = -g \sin \theta$$

14. Now let's do the inclined plane with friction *in general*. Just like the previous problem, use m for mass, θ for angle, and now use μ as a variable for coefficient of friction. Find an expression for the acceleration of the mass as a function of θ , m , and μ .



	x	y
F_g	$-mg \sin \theta$	$-mg \cos \theta$
F_N	0	F_N
F_f	$+\mu \cdot mg \cos \theta$	0
ΣF	$+\mu \cdot mg \cos \theta - mg \sin \theta$	0

$$F_N = mg \cos \theta$$

$$\mu = \frac{F_f}{F_N}$$

$$F_f = \mu F_N$$

$$F_f = \mu \cdot mg \cos \theta$$

For what angle θ is the acceleration = 0?

$$0 = g(\mu \cos \theta - \sin \theta)$$

$$0 = \mu \cos \theta - \sin \theta$$

$$\frac{\sin \theta}{\cos \theta} = \frac{\mu \cos \theta}{\cos \theta}$$

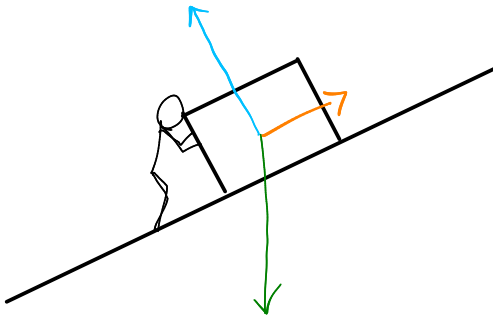
$$\tan \theta = \mu \Rightarrow \theta = \tan^{-1}(\mu)$$

$$ma_x = \Sigma F_x = \mu mg \cos \theta - mg \sin \theta$$

$$\frac{ma_x}{m} = \frac{\mu mg \cos \theta}{m} - \frac{mg \sin \theta}{m}$$

$$a_x = g(\mu \cos \theta - \sin \theta)$$

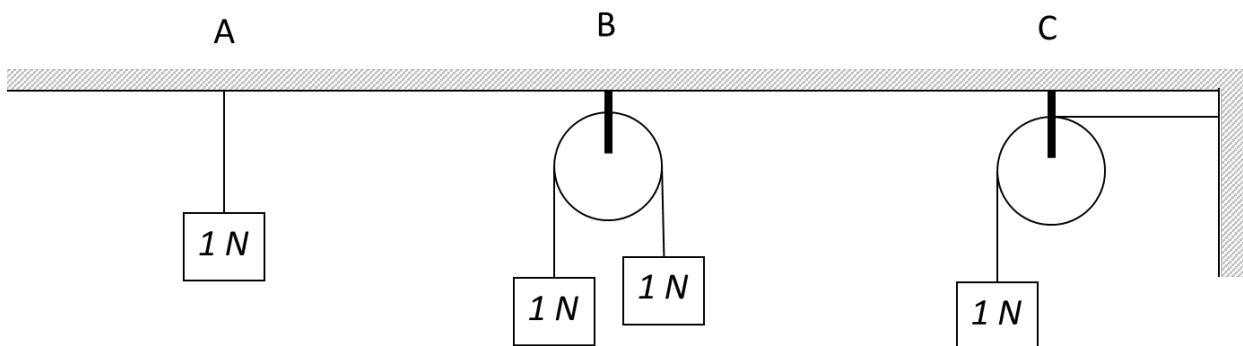
15. In order to *hold* a box on a friction-less inclined plane, that is kept motionless, what force would be necessary to do that? Is there a difference between the force to hold it motionless on the incline and the force to push it up the incline *at constant speed*? No\



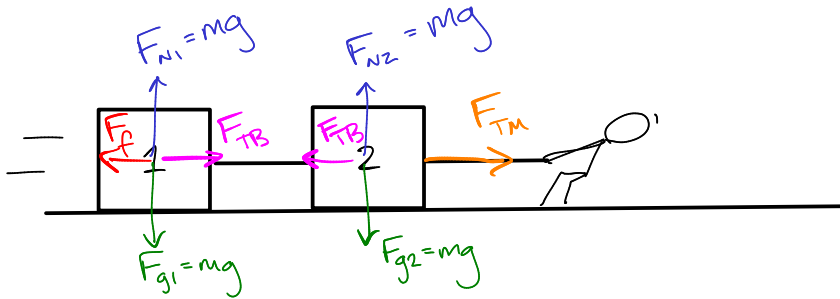
	x	y
F_g	$-mg \sin \theta$	$-mg \cos \theta$
F_N	0	F_N
F_A	F_x	0
ΣF	0	0

$$F_x = mg \sin \theta$$

16. A 50 kg weight is suspended in three different ways shown below. What is the tension in each rope?

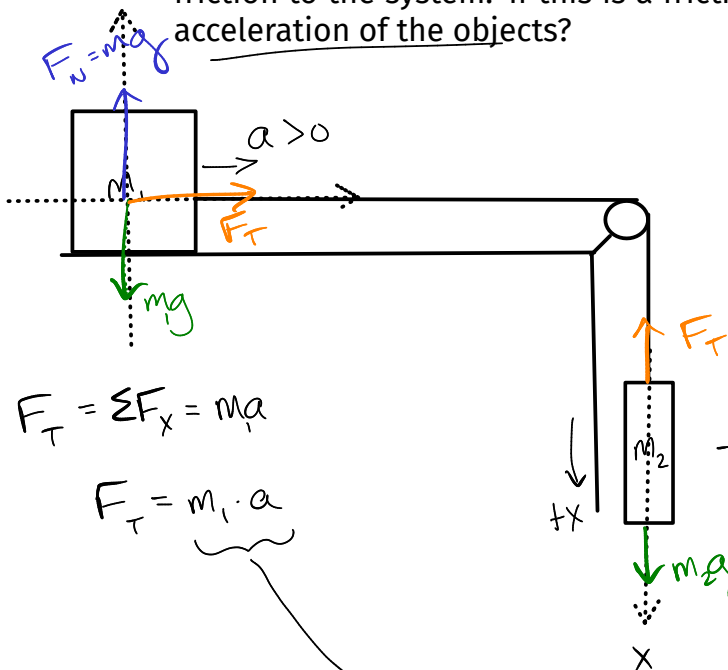


17. I am pulling two objects connected by a string. They are both 100 kg. There is no friction between the objects and the surface, and they are accelerating at 1 m/s^2 . What is my pulling force and what tension force is between the boxes?



$$\begin{aligned}
 -F_p + F_{TB} &= \Sigma F_{x1} = ma_x \\
 F_{TB} &= 100\text{kg} \cdot 1\text{m/s}^2 \\
 \underline{F_{TB} = 100\text{N}}
 \end{aligned}
 \quad
 \begin{aligned}
 -F_{TB} + F_{TM} &= \Sigma F_{x2} = ma_x \\
 -100\text{N} + F_{TM} &= 100\text{kg} \cdot 1\text{m/s}^2 \\
 \underline{F_{TM} = 200\text{N}}
 \end{aligned}$$

18. One object m_1 connected by a string to another object m_2 that is hanging off the edge of the table. There is a pulley at the edge of the table so the string is not introducing any friction to the system. If this is a friction-less table, then what is an expression for the acceleration of the objects?



$$F_T = \Sigma F_x = m_1 a$$

$$F_T = m_1 \cdot a$$

$$-F_T + m_2 g = \Sigma F_x = m_2 a$$

$$-F_T + m_2 g = m_2 a$$

$$-m_1 a + m_2 g = m_2 a$$

$$m_2 g = m_2 a + m_1 a$$

$$m_2 g = (m_1 + m_2) a$$

$$a = \frac{m_2 g}{m_1 + m_2}$$

19. Now do the previous problem again, but with a coefficient of friction μ between the sliding mass (m_1) and the surface.

do for lab

Quiz Q:

